

Research concept note/draft proposal – Jovita Nagawa

Association of U3 Transcription Factor Binding Site Abundance and Genetic Architecture with HIV-1 Latent Viral Reservoir Reactivation in subtypes A1, D and Recombinants.

ABSTRACT

Background: Human immunodeficiency virus type 1 (HIV-1) persists during antiretroviral therapy (ART) through establishment of transcriptionally silent but replication-competent proviruses which is the major barrier to HIV-1 cure (Churchill et al., 2016; Siliciano & Siliciano, 2022). Strategies aimed at reversing HIV latency seek to induce viral transcription and expression (Maina et al., 2021) however, clinical and ex vivo studies have demonstrated that only a fraction of intact proviruses respond to stimulation (Ho et al., 2013, Siliciano & Siliciano, 2021, Khanal et al., 2021). HIV reservoir reactivation is driven by host transcription factors binding to cis-regulatory motifs concentrated within the U3 region of the HIV-1 long terminal repeat (LTR), however, these binding sites remain poorly characterized across HIV-1 subtypes A1, D, and recombinant forms prevalent in Uganda and East Africa (Mbondji-Wonje et al., 2018).

Objectives: This study aims to: (1) systematically locate, quantify, and compare TFBS abundance and sequence architecture for key TFs (NF- κ B, SP1, NFAT, AP-1, TBP, and TFIID) across subtypes A1, D, and recombinants using intact near-full-length proviral sequences; (2) quantify predicted binding score variations attributable to single nucleotide polymorphisms (SNPs) and copy number differences; and (3) correlate U3 TFBS profiles with empirical basal transcription and post-stimulation LVR reactivation outcomes.

Methodology: A cross-sectional analytical study will be conducted using near-full-length proviral sequences generated from Full-Length Intact Proviral Sequencing (FLIP-SEQ) and the HIV Single Molecule Real-Time Capture Assay (HIV SMRTcap) and metadata of the Rakai HIV Reservoir Cohort study. U3 region Transcription Factor Binding Sites (TFBS) will be computationally mapped using JASPAR position weight matrices (PWMs) and the FIMO tool. Motif profiles will be correlated with matched HIV-1 transcription profiling data and Quantitative Viral Outgrowth Assay (QVOA) reactivation metrics using generalised linear and mixed-effects regression models, adjusting for duration on ART, CD4⁺ T-cell count, and viral subtype.

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Introduction

The HIV latent viral reservoir (LVR) refers to long-lived resting CD4⁺ T cells that harbour transcriptionally silent but replication-competent integrated HIV-1 proviruses. These cells are immunologically indistinguishable from uninfected cells, rendering them invisible to the host immune system and insensitive to Combination Antiretroviral Therapy (cART), which exclusively targets active viral replication (Khanal et al., 2021; Siliciano & Siliciano, 2022). Latency reversal is defined as the pharmacological or immunological induction of viral transcription and protein production from these inactive proviruses. The long terminal repeat (LTR) serves as the predominant regulatory element driving viral transcription and is structured into three sub-regions: U3 (unique 3' segment, containing regulatory elements), R (repeated region, including the trans-activation response element, TAR), and U5 (unique 5' segment). The U3 sub-region is further subdivided into the modulatory region (–454 to –104 nt), core enhancer (–105 to –79 nt), and core promoter (–78 to –1 nt), each harbouring binding sites for specific host transcription factors (Pereira et al., 2000; Delannoy et al., 2019).

HIV-1 remains one of the foremost public health epidemics, with 39.9 million people living with the virus globally in 2023 (UNAIDS, 2023). Sub-Saharan Africa accounts for approximately 53% of all people living with HIV despite comprising roughly 10% of the global population (Horvath, 2024). Uganda bears a significant share of this burden and is characterised by an HIV epidemic predominantly driven by subtypes A1, D, and circulating or unique recombinant forms (CRFs/URFs) rather than the globally common subtype C or the well-studied subtype B (Prodger et al., 2017; Kankaka et al., 2023). Despite decades of cART, HIV-1 infection remains incurable because the LVR persists for the lifetime of the host; modelling estimates indicate that even with effective ART-mediated suppression, clearance of the reservoir would require more than 70 years (Siliciano & Siliciano, 2022; McMyn et al., 2023).

Viral transcription is the proximal determinant of latency and reactivation. Under resting conditions (latency), the HIV-1 LTR is kept transcriptionally silent through limited nuclear availability of activating transcription factors, repressive chromatin organisation, and epigenetic silencing (Delannoy et al., 2019; Rodari et al., 2021). Upon host cell stimulation via T-cell receptor (TCR) signalling or cytokine exposure, downstream kinase cascades drive nuclear translocation of key host TFs most notably NF- κ B, NFAT, and AP-1 which bind cooperatively to U3 cis-regulatory elements, recruit RNA polymerase II, and initiate high-velocity viral transcription (Pereira et al., 2000; Burnett et al., 2010). The basal transcription

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machinery components TBP and TFIID anchor the pre-initiation complex at the TATA box within the core promoter, while three SP1 binding sites in the core promoter sustain basal activity and cooperate with NF- κ B in enhancer-driven activation (Verma et al., 2016; Pal et al., 2022).

The U3 region exhibits substantial inter-clade nucleotide divergence, reaching up to 25% across HIV-1 subtypes (Mbondji-Wonje et al., 2018). Such variation is not biologically neutral: the duplication of the NF- κ B binding site in subtype C viruses confers markedly enhanced transcriptional output and a more stable latent state (Pal et al., 2022; Verma et al., 2016). Furthermore, published Massively Parallel Reporter Assay (MPRA) data demonstrate that natural LTR sequence variants spanning even a few nucleotides can produce large, reproducible differences in transcriptional velocity and promoter drive (Burnett et al., 2010). Foundational studies working with subtype B reference sequences, engineered reporter constructs, and plasma-derived viral RNA have progressively characterised how specific U3 TFBS arrangements govern transcriptional output across HIV-1 subtypes. Pereira et al. (2000) compiled a comprehensive catalogue of cellular TF interactions with the subtype B LTR, establishing NF- κ B, SP1, NFAT, and AP-1 binding sites as the core drivers of HIV-1 promoter activity and providing the reference framework against which all subsequent subtype comparisons have been made. Building on this, Burnett et al. (2010) demonstrated using MPRA that subtypes A and D harbour likely compromised Sp1 site II and Sp1 site III respectively deviations predicted to reduce TF binding affinity and alter the transcriptional output of these clades relative to subtype B. Verma et al. (2016) further showed that in subtype C, the physical proximity and precise orientation between the subtype-specific NF- κ B motif (C- κ B) and the adjacent Sp1III site are indispensable for full promoter activation, with disruption of this spatial relationship severely abrogating gene expression even when all other binding sites remain intact. Extending these observations, Pal et al. (2022) demonstrated that NF- κ B copy number directly and proportionally governs transcriptional strength in subtype C, with higher NF- κ B site counts accelerating both transcriptional activation and latency establishment, establishing a direct dose-response relationship between TFBS abundance and promoter drive. At the inter-clade level, Mbondji-Wonje et al. (2018) catalogued U3 sequence variation across 16 HIV-1 clades and recombinant forms using plasma-derived viral sequences, revealing that subtype D carries distinct C/EBP-like motifs and subtype A1 displays non-canonical AP-1/CREB and reduced USF-1 E-box conservation patterns that diverge substantially from the subtype B reference framework on which most mechanistic latency studies have been based. Collectively, these studies establish that U3 TFBS architecture is both subtype-specific and functionally consequential, yet their reliance on reference strains, engineered constructs, and plasma-derived sequences

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leaves the TFBS landscape of intact proviruses within the latent reservoir of subtypes A1, D, and recombinants largely uncharacterised.

The Rakai Health Sciences Program (RHSP) Latent Viral Reservoir (LVR) Cohort in Uganda has generated a unique paired dataset of near-full-length proviral sequences alongside empirical viral outgrowth data. Prodger et al. (2017) demonstrated that virally suppressed Ugandans carry a median LVR size of 0.36 IUPM approximately three-fold lower than an equivalent American cohort hinting that viral genetic and host factors unique to non-B subtypes meaningfully modulate reservoir biology. Despite this, only a few studies have systematically quantified TFBS abundance and motif sequence architecture across subtypes A1, D, and recombinants, nor correlated these U3 level molecular features with functional QVOA-based reactivation. This study introduces this precise integration, potentially revealing viral genetic predictors of reactivation responsiveness that are critically needed to optimise latency-reversing agent (LRA) strategies for African and globally diverse populations.

Problem statement

Despite recognition that host transcription factor (TF) binding to U3 motifs is the proximal driver of HIV-1 reactivation (Churchill et al., 2016; Siliciano & Siliciano, 2022), these binding sites remain poorly characterized across HIV-1 subtypes A1, D, and recombinant forms prevalent in Uganda and East Africa. Latency reversal agents (LRAs) pursuing the "shock-and-kill" strategy have consistently underperformed in clinical trials, reactivating fewer than 5% of latent proviruses even in combination regimens, largely because the molecular determinants of proviral inducibility remain incompletely defined (Debrabander et al., 2023; Spivak & Planelles, 2018).

Studies examining plasma-derived U3 sequences from subtypes A1, D, and recombinant forms have identified subtype-specific deviations in Sp1, C/EBP, and AP-1/CREB binding site architecture, notably compromised Sp1 site III and distinct NF- κ B enhancer configurations in subtype D, and non-canonical C/EBP and reduced USF-1 E-box conservation in subtype A1 suggesting that these sequence-level differences carry functional consequences for transcriptional regulation (Burnett et al., 2010; Mbondji-Wonje et al., 2018). However, these studies relied on plasma-derived viral RNA and consensus reference sequences, which represent circulating replicating virus rather than primary proviral sequences integrated within resting CD4⁺ T cells, the actual cellular compartment constituting the latent reservoir. Critically, while TFBS patterns have been catalogued in reference and plasma-derived sequences, a systematic

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position weight matrix (PWM)-based quantification of TFBS abundance in intact proviruses from subtypes A1, D, and recombinants, linked to empirical QVOA reactivation outcomes from virally suppressed individuals, is minimally studied. Additionally, it remains unresolved whether TFBS abundance exerts a cumulative, dose-dependent effect on inducible transcription or whether a minimum threshold configuration suffices for reactivation. To close this gap, this study will systematically quantify and characterise U3 TFBS abundance and sequence architecture in intact proviruses derived from the latent reservoir of virally suppressed participants in the Rakai HIV Reservoir Cohort, and directly correlate these profiles with matched transcription and QVOA reactivation data.

Significance

This study will generate a comprehensive, patient-derived TFBS abundance map for HIV-1 subtypes A1, D, and recombinants, filling a critical void in non-B HIV cure research. By linking U3 TFBS architecture to empirical QVOA-derived reactivation metrics from a well-characterised Ugandan cohort, findings will provide actionable mechanistic insights to predict and stratify LVR reactivation potential. Results will directly inform the rational design and optimisation of subtype-inclusive shock-and-kill and block-and-lock therapeutic strategies. All bioinformatic pipelines developed will be deposited as open-source tools on GitHub, benefiting the global retrovirology community.

Research Questions

1. What are the quantitative differences in the frequency, spatial positioning, and sequence composition of key U3 TFBS (NF- κ B, SP1, NFAT, AP-1, TBP, TFIID) among and within HIV-1 subtypes A1, D, and their recombinants?
2. To what extent do sequence variants and copy number differences at U3 TFBS alter predicted transcription factor binding scores across and within HIV-1 subtypes A1, D, and their recombinants?
3. How do U3 TFBS abundance profiles and sequence configurations correlate with baseline viral transcription levels and post-stimulation QVOA reactivation kinetics?

Objectives

General Objective

To evaluate the association between the genetic architecture and abundance of U3 region transcription factor binding sites and the functional reactivation capacity of the latent HIV-1 reservoir across HIV-1

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subtypes A1, D, and recombinants using paired proviral genomic and QVOA reactivation data from the Rakai HIV Reservoir Cohort.

Specific Objectives

1. To extract and annotate U3 sequences intact near-full-length HIV-1 proviral genomes (FLIP-SEQ and HIV SMRTcap) in the Rakai Cohort, stratified by subtype (A1, D, CRF/URF), and to systematically map and quantify binding sites for NF- κ B, SP1, NFAT, AP-1, TBP, and TFIID using JASPAR PWMs and the FIMO tool, producing a comprehensive, per-provirus TFBS abundance matrix.
2. To quantify and compare predicted TF binding affinity score variations attributable to SNPs, insertion-deletions, and copy number differences across TFBS for each U3 motif, producing a within- and between-subtype TFBS sequence variant catalogue.
3. To correlate per-provirus U3 TFBS abundance profiles and sequence architecture scores with matched baseline viral transcription and post-stimulation QVOA p24 outgrowth kinetics from the Rakai HIV Reservoir Cohort, using adjusted generalized linear regression models, and to determine whether TFBS abundance exerts a cumulative or threshold effect on reactivation outcomes.

Methodology

Study Design and data acquisition

This is a cross-sectional analytical study utilizing existing prospectively collected clinical, virological, and genomic data from the Rakai HIV Reservoir Cohort.

Study Site and Setting

The study will be conducted at African Center of Excellence in Bioinformatics and Intensive data sciences- Makerere University in collaboration with the Rakai Health Sciences Program (RHSP), Kalisizo, Rakai District, Uganda and University of Western Ontario, Canada.

Study Population

The source population is the RHSP Latent Viral Reservoir (LVR) Cohort, comprising 90 HIV-1-infected adults (≥ 18 years) who have maintained viral suppression on ART for at least one year (two plasma HIV-1 RNA measurements of < 40 copies/mL, 10–18 months apart, with no intervening detectable result), predominantly infected with HIV-1 subtypes A1, D, and recombinants (Prodger et al., 2020; Mumby et al., 2025). The target study population is a subset of cohort participants for whom paired near-full-length

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proviral sequences (FLIP-SEQ or HIV SMRTcap), HIV-1 transcription profiling data, and QVOA reactivation data are available.

Sample Size

The analytic sample will include all cohort participants (~40 individuals) for whom at least one intact proviral sequence, a matched HIV transcription profile, and a QVOA measurement are simultaneously available. This is estimated at 40–70 participants, providing sufficient statistical power (80%, $\alpha = 0.05$) to detect a moderate correlation ($r \geq 0.35$) between TFBS abundance and QVOA-derived reactivation outcomes using two-tailed Pearson/Spearman tests and adjusted regression.

Sampling Technique

A purposive (criterion-based) sampling strategy will be applied, selecting all eligible individuals from the RHSP LVR Cohort who simultaneously satisfy the inclusion criteria outlined below. No probability sampling will be applied, as this study uses a fully pre-existing dataset.

Inclusion and Exclusion Criteria

Inclusion criteria: (1) RHSP LVR Cohort participant aged ≥ 18 years; (2) HIV-1 positive and on suppressive ART with documented viral suppression (<40 copies/mL) for ≥ 1 year; (3) availability of ≥ 1 intact or near-full-length HIV-1 proviral sequence (FLIP-SEQ or HIV SMRTcap) (4) availability of matched basal HIV-1 transcription profiling data and/or QVOA outgrowth data; and (5) confirmed HIV-1 subtype classification as A1, D, or CRF/URF.

Exclusion criteria: (1) Participants with incomplete or failed U3 sequence assembly; (2) sequences classified as subtype B or C; (3) participants with documented ART treatment failure (viral load >400 copies/mL) within 6 months before genomic sampling; and (4) proviral sequences with ambiguous clade classification precluding subtype assignment.

Data Acquisition

Near-full-length proviral sequence data (FASTQ files) and associated participant metadata, including QVOA reactivation outcomes and transcription profiling results, will be obtained from the Rakai Health Sciences Program (RHSP) Latent Viral Reservoir Cohort. Access to these de-identified datasets will be granted through the established RHSP data request protocol, which requires submission of a study proposal for review and approval by the relevant data access committee prior to release.

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Data Processing and Analysis

Bioinformatic analysis will proceed in three sequential phases:

Phase 1 - U3 Sequence Extraction and Subtyping: Full-length proviral sequences will be processed to isolate 5' LTR U3 domains (nucleotides –454 to –1 relative to the transcription start site). Subtype classification will be validated using the Los Alamos Proviral Sequence Database COMET and automated phylogenetic alignment tools.

Phase 2 -TFBS Mapping and Abundance Quantification: Extracted U3 sequences will be scanned using the FIMO tool (MEME Suite) with JASPAR 2024 PWMs for NF- κ B (p50/p65), SP1, NFAT, AP-1 (FOS/JUN), TBP, and TFIID subunits. Output metrics will include: TFBS count per sequence, genomic coordinates, binding affinity scores (log-likelihood ratio scores), and SNP-associated score deviations. Standardised, reproducible R/Python/Bash scripts will be developed for all analyses and deposited on GitHub.

Phase 3 - Statistical Correlation and Regression: TFBS abundance profiles will be correlated with QVOA infectious units per million (IUPM) and baseline transcription levels using Spearman rank correlation. Multivariate generalised linear models (GLMs) and mixed-effects models will assess independent associations, adjusting for ART duration, baseline CD4⁺ T-cell count, pre-ART viral load set-point, sex, and HIV-1 subtype. Interaction terms will test whether subtype modifies the TFBS reactivation association.

Data Cleaning

All sequence data will undergo standard quality control procedures: minimum sequence length thresholds, removal of hypermutated sequences (using the Los Alamos HYPERMUT tool), and exclusion of sequences with >5% ambiguous nucleotide calls in the U3 region. Clinical metadata will be cross-validated against RHSP primary records for completeness, and missing data managed using complete-case analysis or multiple imputation where appropriate.

Data Security

All participant-linked data will be accessed only by authorised investigators through RHSP's secure server infrastructure, consistent with RHSP data governance policies. Genomic sequences will be stored in de-identified form, linked to participant records only through anonymised study IDs. Data transfer between RHSP (Uganda), ACE-Uganda and Western University (Canada) will be encrypted (AES-256) in

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compliance with institutional data security standards. Access logs will be maintained for all data handling activities.

Dissemination

Findings will be disseminated through: (1) an MSc thesis submitted to Makerere University; (2) presentation at the Rakai Health Sciences Program scientific meeting, (3) at least one peer-reviewed manuscript in a high-impact virology or HIV cure journal; (4) presentation at HIV conferences; and (5) deposition of all bioinformatic pipelines and processed data as open-access resources on GitHub and a public genomics repository.

Ethical considerations

This study will use entirely pre-existing, previously consented biological samples and associated data from the RHSP LVR Cohort. Ethical approvals for the original cohort were obtained from the Institutional Review Boards (IRBs) of the Uganda Virus Research Institute (UVRI), the Uganda National Council for Science and Technology (UNCST), the National Institutes of Health (NIH, USA), and Western University (Canada) (Kankaka et al., 2023; Prodger et al., 2017). All original participants provided written informed consent at each study visit for future use of their de-identified data and samples in HIV-related research. An amendment/waiver for secondary use of existing data will be sought from the UVRI and UNCST IRBs prior to study commencement. Additional approval will be obtained from the Makerere University College of Health Sciences School of Biomedical Sciences Research Ethics Committee in fulfillment of the requirements for the conduct of a Master's thesis.

No new biological samples will be collected, and there will be no direct participant contact, negating the need for re-consent. All data will be handled in a de-identified manner consistent with RHSP's institutional data use policies. Privacy and confidentiality will be maintained through strict access controls, anonymised study IDs, and encrypted data storage and transfer protocols. This study presents no greater than minimal risk to participants. Findings will be reported in aggregate without participant-identifiable information.

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References

1. Burnett, J. C., Lim, K. I., Calafi, A., Rossi, J. J., Schaffer, D. V., & Bhatt, A. P. (2010). Combinatorial latency reactivation for HIV-1 subtypes and variants. *Journal of Virology*, 84(12), 5958–5974.
2. Debrabander, Q., Bontems, S., Koch, A., & Darcis, G. (2023). The efficacy and tolerability of latency-reversing agents in reactivating the HIV-1 reservoir in clinical trials. *Journal of Virus Eradication*, 9(2), 100345.
3. Delannoy, A., Poirier, M., & Bell, B. (2019). Cat and mouse: HIV transcription in latency, immune evasion, and cure/remission strategies. *Viruses*, 11(3), 269.
4. Fornes, O., Castro-Mondragon, J. A., Khan, A., van der Lee, R., Zhang, X., Richmond, P. A., ... & Sandelin, A. (2020). JASPAR 2020: Update of the open-access database of transcription factor binding profiles. *Nucleic Acids Research*, 48(D1), D87–D92.
5. Kankaka, E. N., Redd, A. D., Khan, A., Reynolds, S. J., Saraf, S., Kirby, C., ... & Poon, A. (2023). Dating reservoir formation in virologically suppressed people living with HIV-1 in Rakai, Uganda. *Virus Evolution*, 9(2), vead046. <https://doi.org/10.1093/ve/vead046>
6. Khanal, S., Schank, M., El Gazzar, M., Moorman, J. P., & Yao, Z. Q. (2021). HIV-1 latency and viral reservoirs: Existing reversal approaches and potential technologies, targets and pathways. *Vaccines*, 9(5), 1110.
7. Mbondji-Wonje, C., Dong, M., Wang, X., Zhao, J., Ragupathy, V., Sanchez, A. M., ... & Hewlett, I. (2018). Distinctive variation in the U3R region of the 5' long terminal repeat from diverse HIV-1 strains. *PLOS ONE*, 13(4), e0195661.
8. McMyn, N. F., Varriale, J., Bhatt, N. P., Ott, S., Marsden, M., Dukhovlinova, E., ... & Margolis, D. M. (2023). The latent reservoir of inducible, infectious HIV-1 does not decrease despite decades of antiretroviral therapy. *Journal of Clinical Investigation*, 133(14), e171554.
9. Mumby, M., Kankaka, E., Khan, A., Redd, A. D., Quinn, T. C., Galiwango, R. M., ... & Dikeakos, J. D. (2025). Association between HIV-1 Nef-mediated MHC-I downregulation and the maintenance of the replication-competent latent viral reservoir in Uganda. *Lancet Microbe*, 6, 101044.
10. Pal, S., Jaiswal, V., Nala, N., & Ranga, U. (2022). Enhanced transcriptional strength of HIV-1 subtype C minimizes gene expression noise and confers stability to the viral latent state. *Journal of Virology*, 97(1), e01376–22.

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11. Pereira, L. A., Bentley, K., Peeters, A., Churchill, M. J., & Deacon, N. J. (2000). A compilation of cellular transcription factor interactions with the HIV-1 LTR promoter. *Nucleic Acids Research*, 28(3), 663–668.
12. Prodger, J. L., Lai, J., Reynolds, S. J., Keruly, J. C., Moore, R. D., Kasule, J., ... & Quinn, T. C. (2017). Reduced frequency of cells latently infected with replication-competent HIV-1 in virally suppressed individuals living in Rakai, Uganda. *Clinical Infectious Diseases*, 65(8), 1308–1315.
13. Prodger, J. L., Capoferri, A. A., Yu, K., Lai, J., Reynolds, S. J., Kasule, J., ... & Quinn, T. C. (2020). Reduced HIV-1 latent reservoir outgrowth and distinct immune correlates among women in Rakai, Uganda. *JCI Insight*, 5(14), e139287.
14. Rodari, A., Darcis, G., & Van Lint, C. M. (2021). The current status of latency reversing agents for HIV-1 remission. *Annual Review of Virology*, 8, 491–514.
15. Sadri, G., Nadakal, T. T., Lauer, W., Kos, J., Singh, P. K., Elliott, E., ... & Prodger, J. L. (2026). Development and validation of HIV SMRTcap for the characterisation of HIV-1 reservoirs across tissues and subtypes. *PLOS Pathogens*, 22(1), e1013171.
16. Siliciano, J. D., & Siliciano, R. F. (2022). In vivo dynamics of the latent reservoir for HIV-1: New insights and implications for cure. *Annual Review of Pathology*, 17, 271–294.
17. Spivak, A. M., & Planelles, V. (2018). Novel latency reversal agents for HIV-1 cure. *Annual Review of Medicine*, 69, 421–436.
18. Verma, A., Rajagopalan, P., Lotke, R., Varghese, R., Selvam, D., Kundu, T. K., & Ranga, U. (2016). Functional incompatibility between the generic NF- κ B motif and a subtype-specific Sp1III element drives the formation of the HIV-1 subtype C viral promoter. *Journal of Virology*, 90(16), 7046–7065.